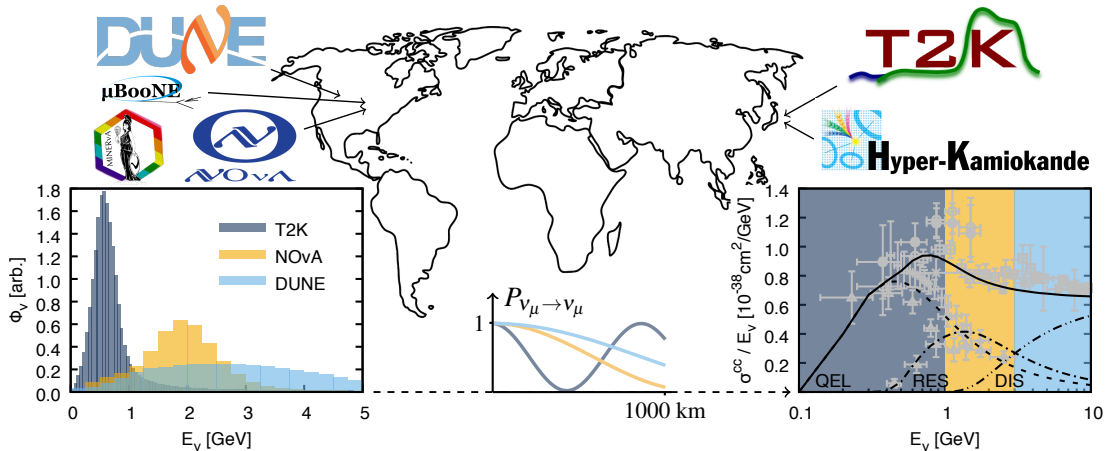


Physics Landscape of NuWro

Kajetan Niewczas





$$P(\nu_\mu \rightarrow \nu_e) \simeq \sin^2(2\theta) \sin^2 \left(1.27 \frac{\Delta m^2 L}{E_\nu} \right)$$

↑
oscillation

↑
amplitude

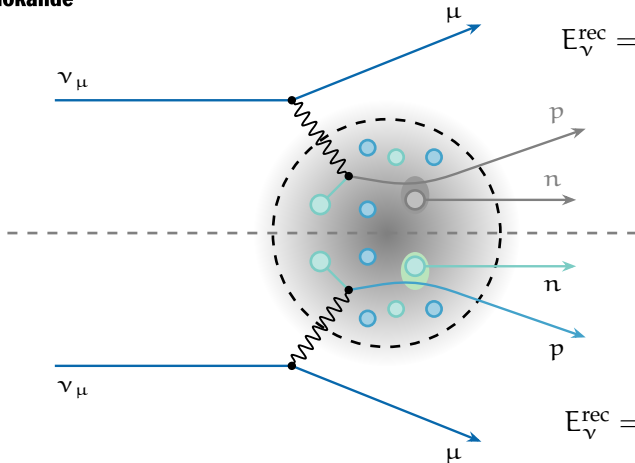
↑
frequency

$$A_{CP} = \frac{P(\nu_\mu \rightarrow \nu_e) - P(\bar{\nu}_\mu \rightarrow \bar{\nu}_e)}{P(\nu_\mu \rightarrow \nu_e) + P(\bar{\nu}_\mu \rightarrow \bar{\nu}_e)}$$

↑
asymmetry

↑
oscillation ratio

Kinematical energy reconstruction



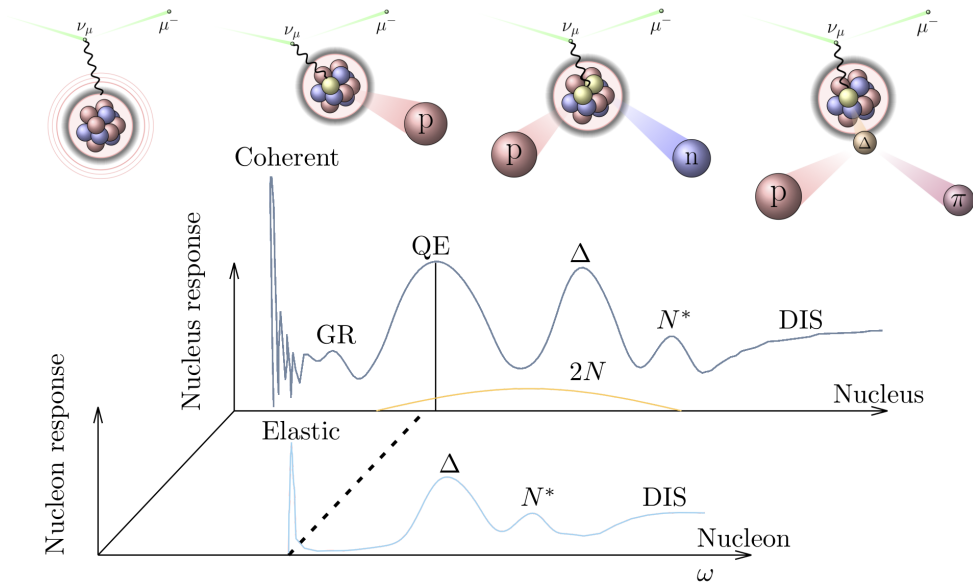
$$E_{\nu}^{\text{rec}} = \frac{2M_N E_{\mu} - m_{\mu}^2 + M_{N'}^2 - M_N^2}{2(M_N - E_{\mu} + p_{\mu} \cos \theta)}$$

$$E_{\nu}^{\text{rec}} = E_{\mu} - E_B + \sum_{\text{nuc.}} T_i + \sum_{\text{mes.}} E_j$$



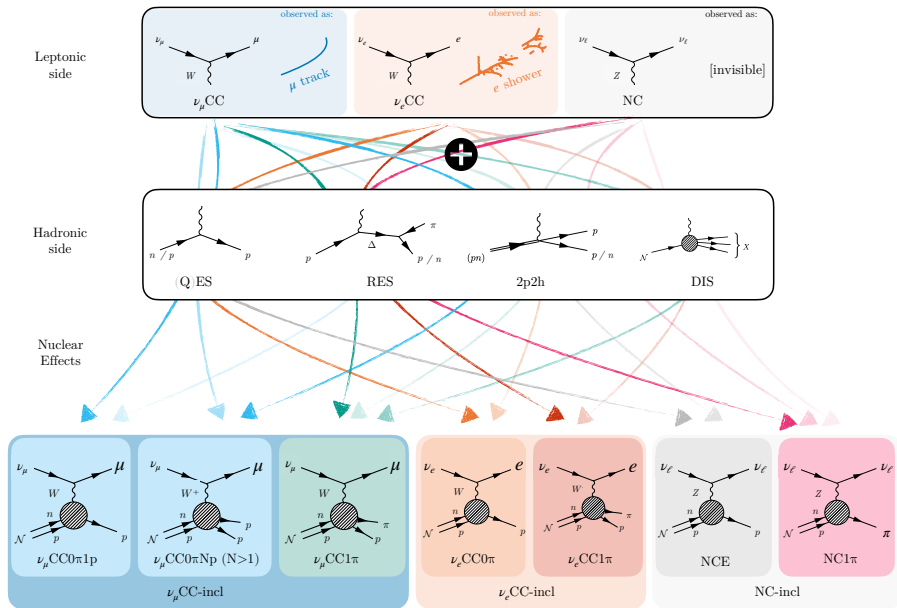
Calorimetric energy reconstruction

Nuclear response



Credit:
Tom van Cuyck

Experimental topologies

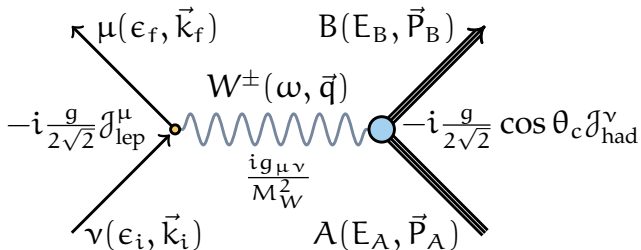


Credit:
Katharina Lachner

Event generation and essential approximations

What does it mean to generate an event?

- we need to fix the **four-momenta** for all **final-state particles**
- a set of **independent variables** needs to respect the probability distribution set by the **differential cross section**



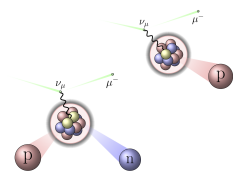
$$P(c, x_c | E_\nu) = P(c | E_\nu) \cdot P(x_c | c, E_\nu) = \frac{\sigma_c(E_\nu)}{\sigma_{\text{tot}}(E_\nu)} \cdot \frac{1}{\sigma_c(E_\nu)} \frac{d^{(N_c)} \sigma_c(E_\nu)}{dx_{c,1} \cdots dx_{c,N_c}}$$

→ where c is a channel and N_c is the dimensionality of a given channel

- **neutrino energy** is a **fixed input**, for realistic beams we perform flux folding:

$$\langle P(c, x_c) \rangle_\Phi = \int dE_\nu \Phi(E_\nu) P(c, x_c | E_\nu)$$

Degrees of freedom



Unknown particle 4-vectors	Variables	Physical effects	Variables
Initial lepton	4	Particles on-shell	$-(3 + N)$
Target nucleus	4	4-momentum conservation	-4
Final lepton	4	Target rest-frame	-3
Remnant nucleus	4	Fixed projectile direction	-2
Outgoing hadrons	$4N$	Fixed incoming energy	-1
	$16 + 4N$		$-13 - N$
			$3 + 3N$

- Final-state lepton:

$$d^3\vec{k}' \rightarrow dE_{k'} d^2\Omega_{k'} \rightarrow dE_{k'} d\cos\theta_{k'} d\phi_{k'} \rightarrow d\omega d|\vec{q}| d\phi_{k'} \rightarrow dW dQ^2 d\phi_{k'}$$

→ with a **rotational symmetry** in $\phi_{k'}$

- Final-state hadrons:

$$d^3\vec{p}' \rightarrow dE_p d^2\Omega_{p'} \rightarrow dT_p d\cos\theta_p d\phi_{k'p'}$$

→ keeping the **sensitivity** to the relative $\phi_{k'p'}$ angle

- from $3N + 3$: **-3** if the target is a **free nucleon**; **-1** if the scattering is (quasi)**elastic** (fixed W)

Modeling approximations

$$|\mathcal{M}|^2 \propto |\langle \Psi_A | i\hat{T} | X \rangle|^2$$

Born approximation

CRPA

$$\approx \left| \sum_{\alpha} \langle \Psi_A | i\hat{T} | \Psi_{\alpha} \rangle \langle \Psi_{\alpha} | X \rangle \right|^2 \approx \sum_{\alpha} |\langle \Psi_A | i\hat{T} | \Psi_{\alpha} \rangle|^2 |\langle \Psi_{\alpha} | X \rangle|^2$$

+ intermediate hadronic state + time separation of FSI (MC factorization)

DWIA

$$\approx \sum_{\alpha, h, p} \left| \langle \psi_h | \hat{O}_{1b} | \psi_p \rangle \langle \Psi_A | \hat{a}_h^{\dagger} \hat{a}_p | \Psi_{\alpha} \rangle \right|^2 |\langle \Psi_{\alpha} | X \rangle|^2$$

+ one-body nuclear currents

PWIA

$$\approx \sum_{h, p} |\langle \psi_h | \hat{O}_{1b} | \psi_p \rangle|^2 \left| \langle \Psi_A | \hat{a}_h^{\dagger} | \Psi_{A-1} \rangle \right|^2 |\langle \Psi_{A-1}, \psi_p | X \rangle|^2$$

+ cross section factorization (PWIA)

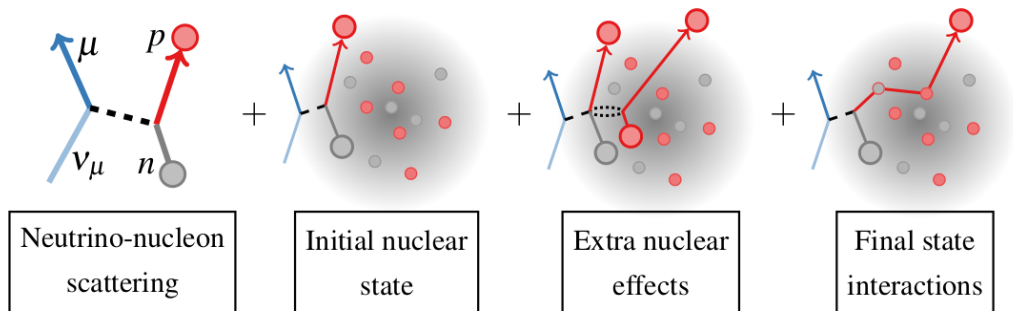
Generators

$$\approx \sum_{h, p} \sigma_{hp} S_h(E, p) P(p|X)$$

+ intranuclear cascade

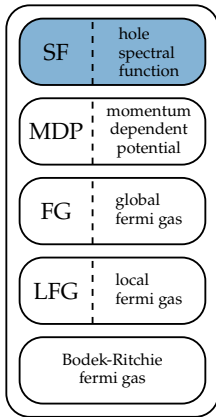
KN, MSc
A. Nikolakopoulos et al., Phys.Rev.C 110 (2024) 054611

Monte Carlo event generators

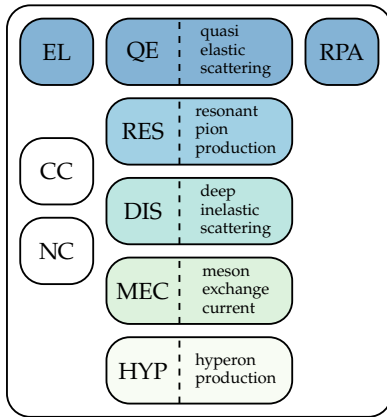


→ Cross sections are evaluated in a **factorized scheme**

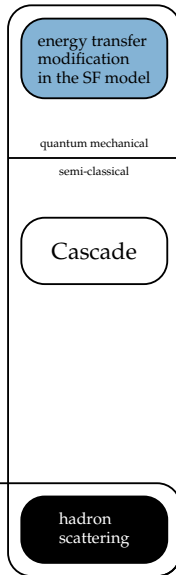
Initial state



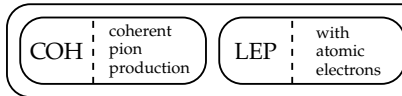
Interaction channel



FSI



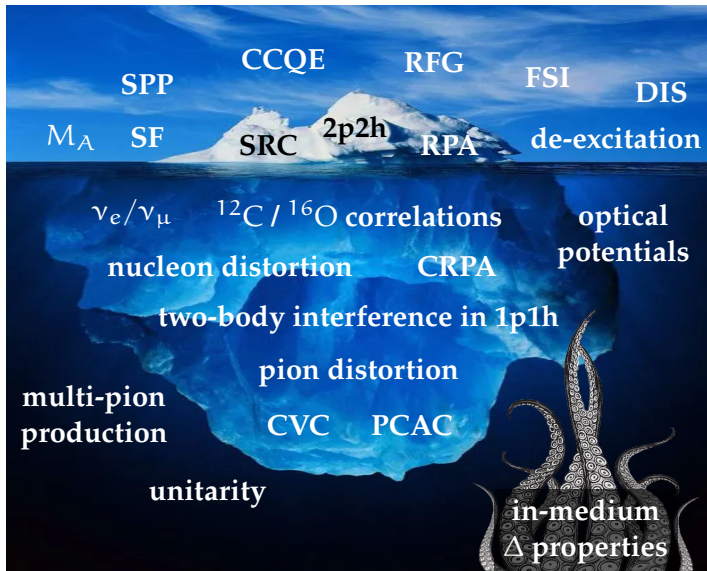
Other interactions



Next-generation experiments



Near-detector constraints



Scattering off a free nucleon

Quasielastic scattering

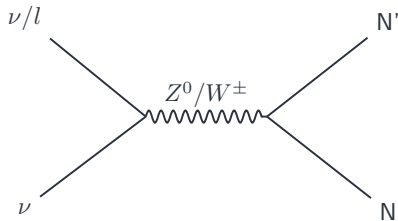
Llewellyn-Smith formula

$$\frac{d\sigma}{d|q^2|} \left(\begin{array}{l} \nu_l + n \rightarrow l^- + p \\ \bar{\nu}_l + p \rightarrow l^+ + n \end{array} \right) = \frac{M^2 G_F^2 \cos^2 \theta_C}{8\pi E_\nu^2} \left[A(Q^2) \mp B(Q^2) \frac{(s-u)}{M^2} + C(Q^2) \frac{(s-u)^2}{M^4} \right]$$

where functions A , C contain the VV and AA terms, while B is purely VA

Default parametrization:

- **Isovector**: $F_1^V(Q^2), F_2^V(Q^2)$
→ BBBA05; *Nucl.Phys.B Proc.Suppl.* 159 (2006) 127-132
- **Axial-vector**: $G_A(Q^2)$
→ MINERvA; *Nature* 614 (2023) 48-53
- **Pseudoscalar**: $F_P(Q^2)$
→ $4M^2/(m_\pi^2 + Q^2)G_A(Q^2)$; PCAC



Inelastic scattering

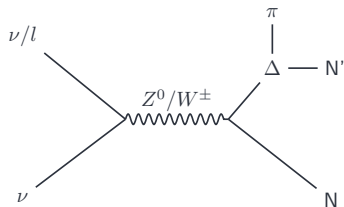
Single-pion production

- Precise models for **resonant pion production**

→ smooth **transition** between **RES** and **DIS** in the single-pion production channel

$$\frac{d\sigma^{\text{RES}}}{dW} = \frac{d\sigma^{\text{SPP}}}{dW} (1 - \alpha(W)) + \frac{d\sigma^{\text{PYTHIA:SPP}}}{dW} F^{\text{SPP}}(W) \alpha(W)$$

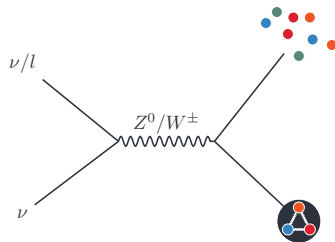
$$F^{\text{SPP}}(W) = \frac{d\sigma^{\text{PYTHIA:SPP}}/dW}{d\sigma^{\text{PYTHIA}}/dW}$$



Multi-pion and beyond

- DIS+PYTHIA to model **particle multiplicity**

$$\alpha(W) = \begin{cases} \frac{W - W_{\text{th}}}{W_{\text{min}} - W_{\text{th}}} \alpha_0 & W < W_{\text{min}} \\ \frac{W - W_{\text{min}} + \alpha_0 (W_{\text{max}} - W)}{W_{\text{max}} - W_{\text{min}}} & W_{\text{min}} \leq W \leq W_{\text{max}} \\ 1 & W_{\text{max}} < W \end{cases}$$



Single-pion production

(first implementation: KN et al., *Phys.Rev.D* 103 (2021) 053003)

Δ -resonance only

- Δ excitation on the nucleon (2d)

$$\left. \frac{d^4\sigma}{dWdQ^2d\Omega_\pi^*} \right|_\Delta = \frac{d^2\sigma}{dWdQ^2} \cdot \frac{1}{2\pi} \mathcal{A}(\cos\theta_\pi^*; Q^2)$$

- Angular distribution fit to ANL or BNL data

Ghent Hybrid model

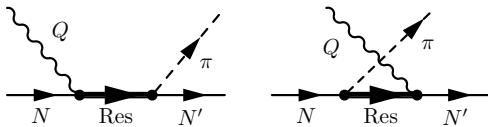
- Full **4-dimensional** cross section

$$\left. \frac{d^4\sigma}{dWdQ^2d\Omega_\pi^*} \right|_{\text{Hybrid}}$$

- RGJ et al., *Phys.Rev.D* 95 (2017) 113007

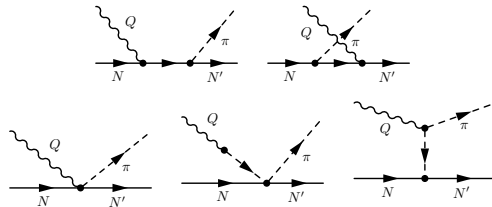
Resonances

$P_{33}(1232)$, $P_{11}(1440)$, $D_{13}(1520)$, $S_{11}(1535)$



based on [PRD 76 033005, PRD 87 113009, PRD 93 014016]

ChPT background



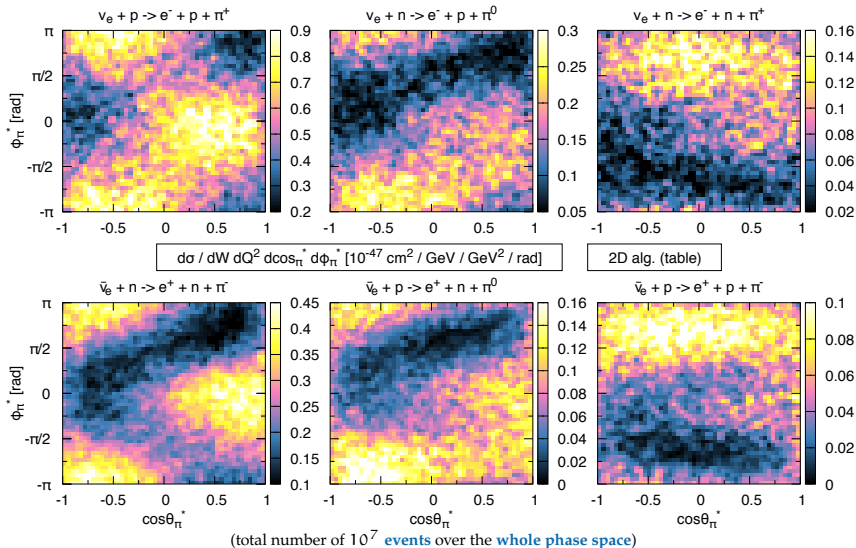
Pion angular distributions

- Ghent LEM
- Free nucleon
- Fixed kinematics:

$$E = 1 \text{ GeV}$$

$$Q^2 = 0.1 \text{ GeV}^2$$

$$W = 1230 \text{ MeV}$$



KN et al.,
Phys.Rev.D 103 (2021) 053003

Deep-inelastic scattering

Inclusive cross section

$$\frac{d^2\sigma^{\nu(\bar{\nu})}}{dx dy} = \frac{G_F^2 ME}{\pi} \left[\left(xy^2 + \frac{ym^2}{2ME} \right) F_1(x, Q^2) \pm \left(xy - \frac{xy^2}{2} - \frac{ym^2}{4ME} \right) F_3(x, Q^2) \right. \\ \left. + \left(1 - y - \frac{Mxy}{2E} - \left(\frac{m}{2E} \right)^2 - \frac{m^2}{2ME_x} \right) F_2(x, Q^2) \right]$$

where $x = Q^2/2M\omega$, $y = \omega/E_\nu$

- The **structure functions** are given by:

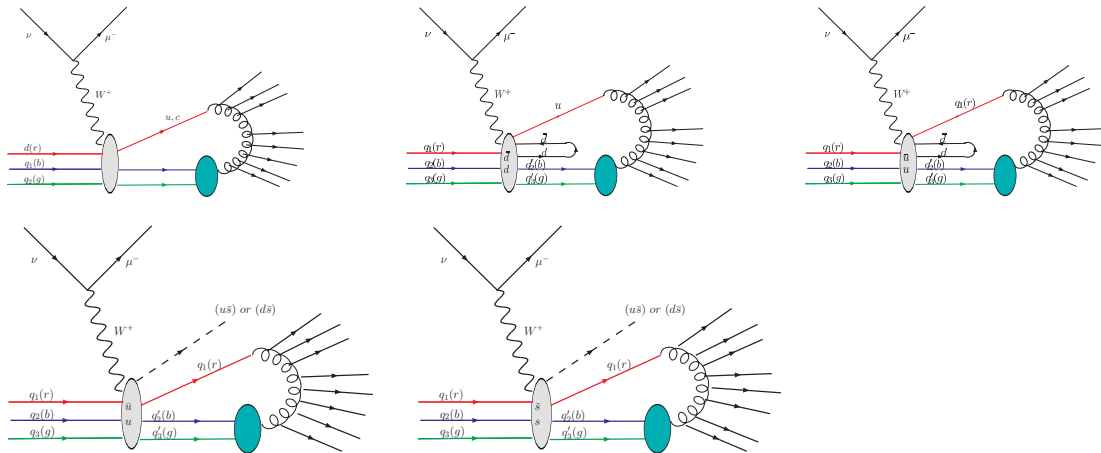
$$F_1(x, Q^2) = \sum_j [q_j(x, Q^2) + \bar{q}_j(x, Q^2)] \quad F_2(x, Q^2) = 2xF_1(x, Q^2)$$

$$F_3(x, Q^2) = 2x \sum_j [q_j(x, Q^2) - \bar{q}_j(x, Q^2)]$$

- PDFs q_j , $\bar{q}_j$ are taken from *Z.Phys.C* 67 (1995) 433-488,
with **low- Q^2 corrections** by *Nucl.Phys.Proc.Suppl.* 112 (2002) 70-76

Fragmentation

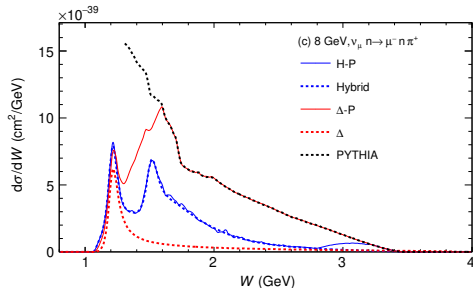
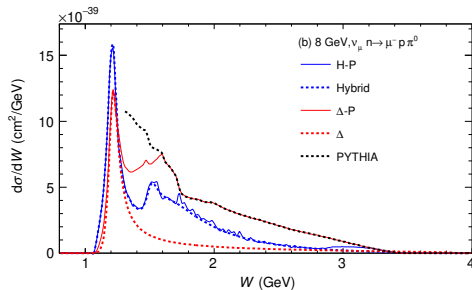
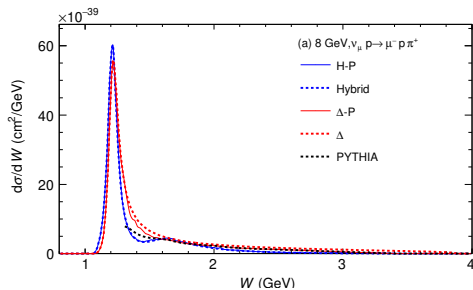
(scenarios depend on the target quark: valence or sea)



→ a **quark** ($q, \bar{q}$) and a **diquark** (qq) form a **string**, which is an input to **PYTHIA6**

Single-pion production

(in these plots, the transition in the H-P model is $W \in (2.8, 3.2)$ GeV)



- **old transition** region: $W \in (1.3, 1.6)$ GeV
 - **new transition** region: $W \in (1.6, 1.9)$ GeV
- new definition of the **RES / DIS boundary**

Qiyu Yan et al., *JHEP* 12 (2024) 141

Hyperon production

- Probing the **strangeness component** of the nucleon:

$$\bar{\nu}_l + p \rightarrow \Lambda + l^+$$

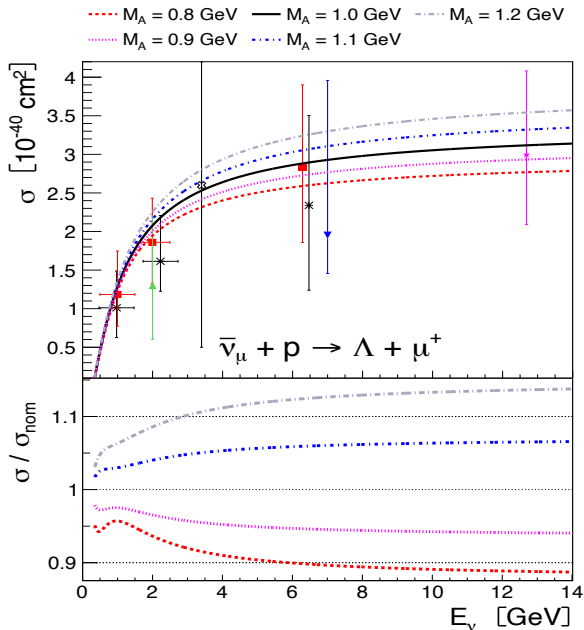
$$\bar{\nu}_l + p \rightarrow \Sigma^0 + l^+$$

$$\bar{\nu}_l + n \rightarrow \Sigma^- + l^+$$

→ plus Σ^+ due to reinteractions in nuclear medium

- Cabibbo-suppressed** ($\Delta S = 1$): small but non-negligible contribution

C. Thorpe et al., *Phys.Rev.C* 104 (2021) 035502



Nuclear models and interaction dynamics

Independent-particle model

Observations:

- nuclei are **made of nucleons**,
- **binding per nucleon** is relatively small ($\simeq 7.5$ MeV for ^{12}C),
- **distance between particles** larger than the nucleon radius ($\simeq 1 - 2$ fm),

Probability for a particle to propagate over a distance x with **no interactions** is

$$P(x) = \exp(-x/\lambda)$$

where $\lambda = (\rho\sigma)^{-1}$ is the **mean free path**, while ρ is **target density** and σ is **interaction cross section**

For nucleons inside nuclei:

$$\tilde{\lambda} \ll d < \lambda < R$$

where $\tilde{\lambda}$ is the de Broglie **wavelength**, d is the **distance** between targets, and R is the **nuclear radius**

→ nucleus can be modeled as a system of **independent, quasifree nucleons**

Essential nuclear models

Fermi gas (**FG**):

$$p_F = \frac{\hbar}{r_0} \left(\frac{9\pi N}{4A} \right)^{1/3} = \hbar \left(3\pi^2 \rho_{\text{avg}} \frac{N}{A} \right)^{1/3}$$

Local fermi gas (**LFG**):

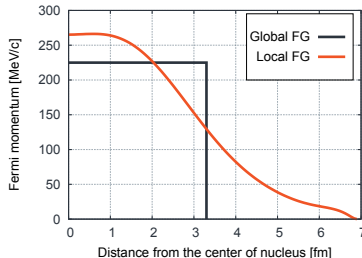
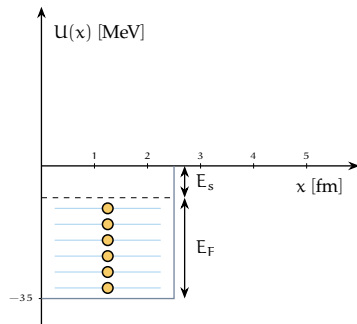
$$p_F(r) = \hbar \left(3\pi^2 \rho(r) \frac{N}{A} \right)^{1/3}$$

→ **nucleon transition** from below the Fermi sea to above

$$\theta(k_F - |\vec{k}|) \rightarrow \theta(|\vec{k} + \vec{q}| - k_F)$$

$$|\vec{k}|^2/2M - E_s - E_F \rightarrow (|\vec{k} + \vec{q}|^2 + M^2)^{1/2}$$

→ final nucleon is always a **plane wave**

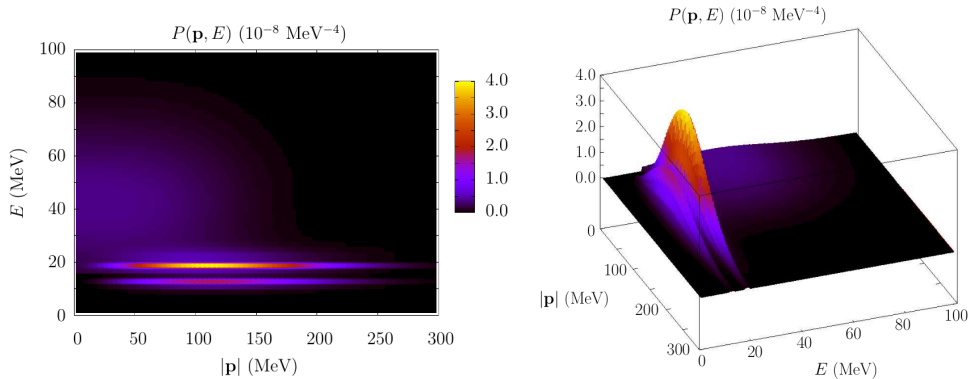


QE: Spectral function

Factorization of the **neutrino-nucleus** cross section in the absence of FSI:

$$\frac{d^2\sigma^{\text{PWIA}}}{d\omega d|\vec{q}|} = \int dE d^3\vec{p} \frac{G_F^2 \cos^2 \theta_C |\vec{q}|}{4\pi E_k^2 E_p E_{p'}} P(E, \vec{p}) L_{\mu\nu} \tilde{H}^{\mu\nu} \delta(\omega + M - E - E_{p'})$$

→ $L_{\mu\nu} \tilde{H}^{\mu\nu}$ is the off-shell **neutrino-nucleon** interaction

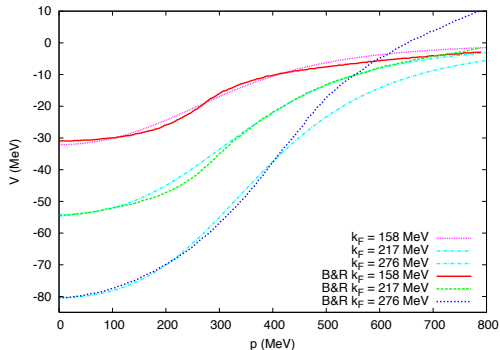


→ for the **latest improvements**, see the **talk by Rwik**

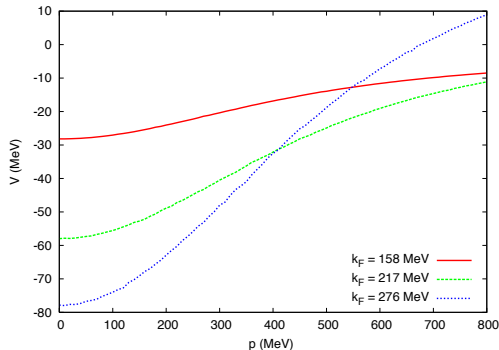
QE: Momentum-dependent potential

- Nucleon **energy** calculated using a **density**- and **momentum**-dependent **potential**

$$E_k + E_p + V(p, k_F) = E_{k'} + E_{p'} + V(p', k_F)$$



left: the Brieva-Dellafiore potential and the approximation used in NuWro



right: the potential used in GiBUU

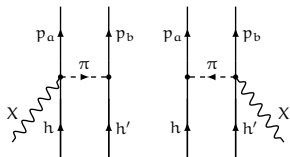
Meson-exchange currents

Explicit **two-body currents** contributing to the **2p2h** final-states:

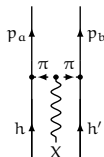
Models available in NuWro:

- Valencia
- Transverse Enhancement Model
- Marteau-like
- SuSAv2

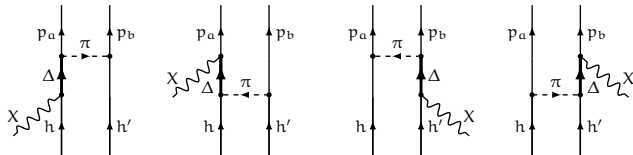
→ **Seagull** currents



→ **Pion-in-flight** current

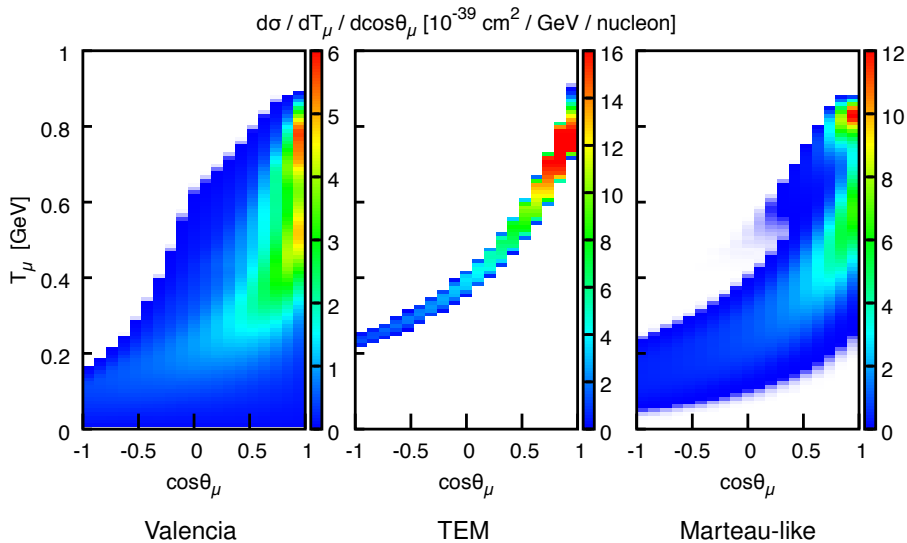


→ **Δ -isobar** degrees of freedom



→ for the **latest updates** to the **Valencia model**, see the **talk by Hemant**

Meson-exchange currents



Scattering of 1 GeV
neutrinos on Carbon

Multinucleon final states

1.



Choose p_1, p_2 back-to-back

2.



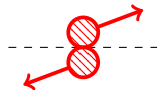
$$(p_1 + p_2 + q)^2 > 4M_N^2$$

3.



Boost to CMS frame

4.



Decay isotropically

5.



Boost back to LAB frame

6.

$$\Theta(k_F - |\vec{p}'_1|)\Theta(k_F - |\vec{p}'_2|)$$

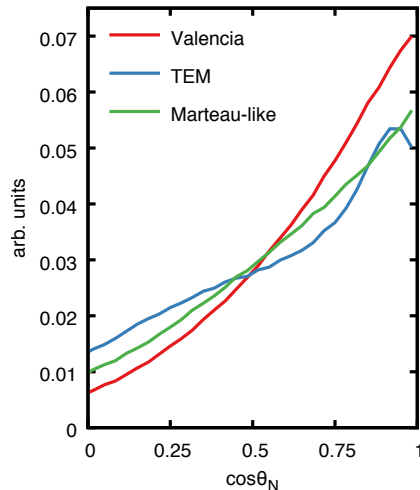
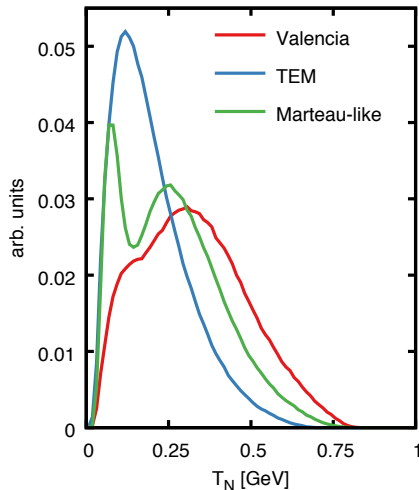
Check Pauli blocking

→ for the
**3-nucleon
knockout**,
see the **talk
by Hemant**

J. Sobczyk, *Phys.Rev.C* 86 (2012) 015504

Multinucleon final states

Distributions for the **most energetic nucleon** before FSI



Scattering of 1 GeV
neutrinos on Carbon

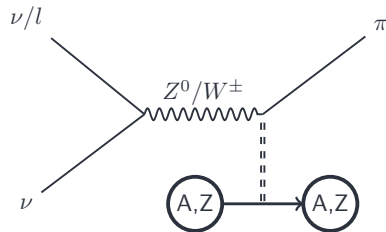
Coherent pion production

Production of pions **without** the nucleus **excitation**:

$$\nu_l + A \rightarrow \nu_l + \pi^0 + A$$

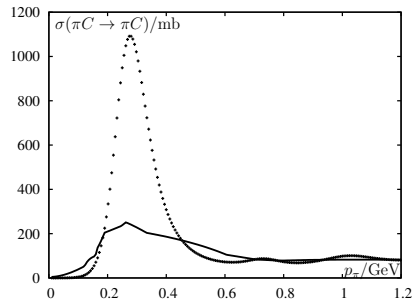
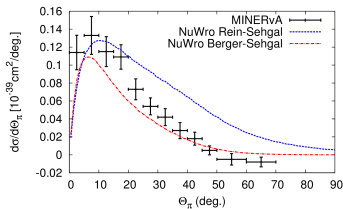
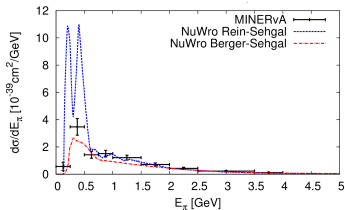
$$\nu_l + A \rightarrow l^- + \pi^+ + A$$

$$\bar{\nu}_l + A \rightarrow l^+ + \pi^- + A$$



Total πA cross sections as input:

- **Rein-Sehgal** model [dotted] → **model dependent**
- **Berger-Sehgal** model [full] → **experimental data**

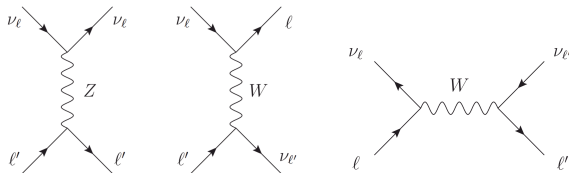


Scattering on atomic electrons

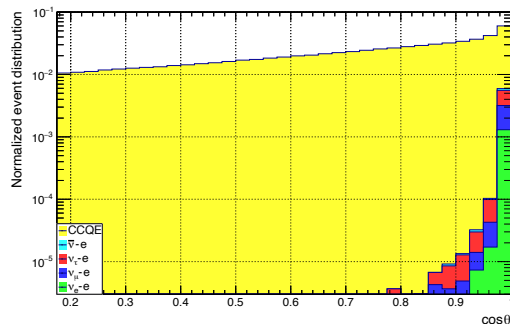
Pure **Standard Model** process (interference of CC and NC)

$$\nu_l + e^- \rightarrow \nu_l + e^- \quad (\bar{\nu}_l + e^- \rightarrow \bar{\nu}_l + e^-)$$

→ no nuclear-model uncertainty, so very useful for **flux calibration**



D. Zhuridov et al., *J.Phys.G* 48 (2021) 055002



Final-state interactions

Optical potential prescription

Effective FSI via **smearing** and **redistributing** the energy transfer:

$$\frac{d\sigma^{\text{FSI}}}{d\omega d\Omega} = \int d\omega' f_{\bar{q}}(\omega - \omega' - U_V) \frac{d\sigma^{\text{IA}}}{d\omega' d\Omega}$$

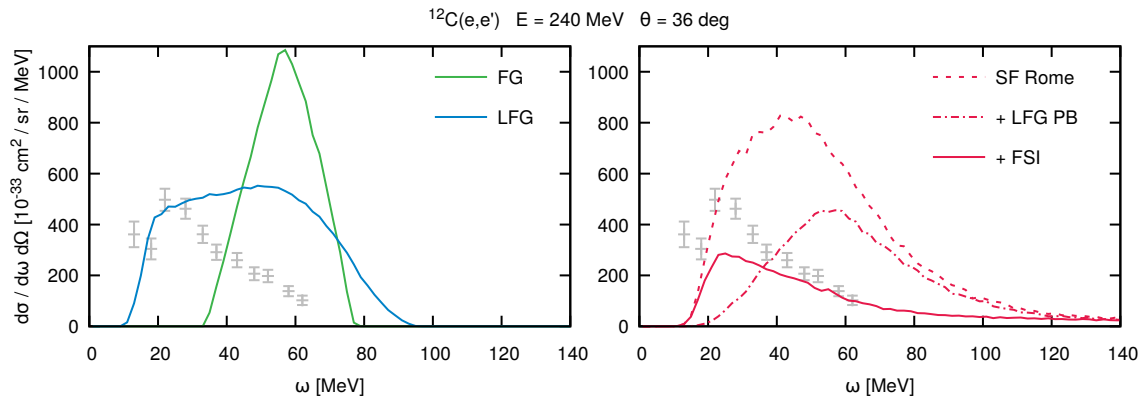
where the **folding function** is

$$f_{\bar{q}}(\omega) = T_A \delta(\omega) + (1 - T_A) F_{\bar{q}}(\omega) \quad F_{\bar{q}}(\omega) = \frac{1}{\pi} \frac{U_W}{U_W^2 + \omega^2}$$

- T_A : **nuclear transparency**
- $U = U_V + iU_W$: **optical potential** (real / imaginary parts)

→ for more information, see the **talk by Rwik**

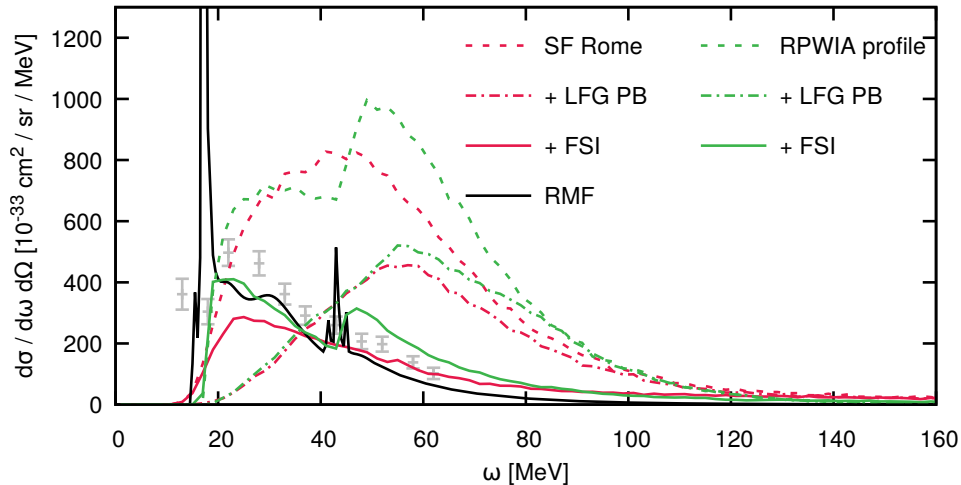
QE: Electron scattering



→ **FG** and **LFG** do not reproduce the inclusive electron results

QE: Electron scattering

$^{12}\text{C}(e,e')$ $E = 240 \text{ MeV}$ $\theta = 36 \text{ deg}$



→ For inclusive cross sections, the **correction** reproduces the data well

Intranuclear cascade

- **Propagates particles** through the nuclear medium
- **Probability** of passing a distance λ :

$$P(\lambda) = e^{-\lambda/\tilde{\lambda}}$$

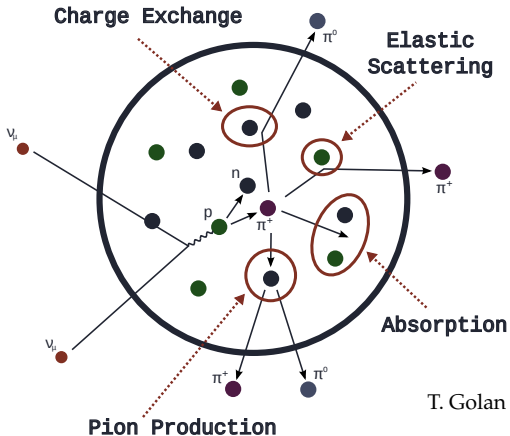
where $\tilde{\lambda} \equiv (\rho\sigma)^{-1}$ and

ρ - local density

σ - cross section

→ Implemented for **nucleons**, **pions** and **hyperons**

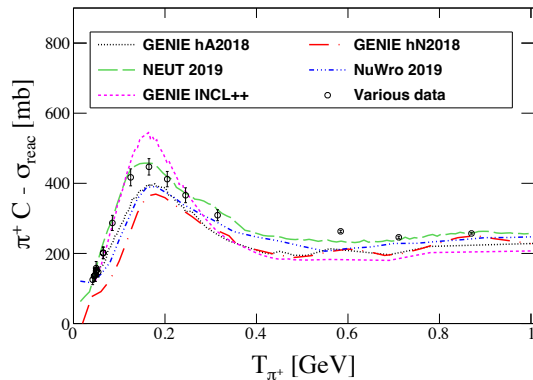
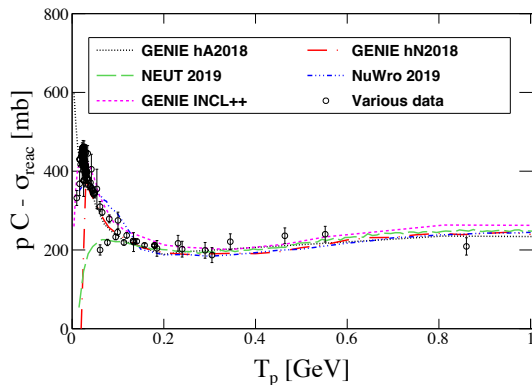
T. Golan, C. Juszczak, J.T. Sobczyk,
Phys.Rev. C 86 (2012) 015505



T. Golan

Standalone cascade

- Provides the ability to calculate **hadron scattering** cross sections
- Allows for many **advanced FSI analyses** and debugging



S. Dytman et al., *Phys.Rev.D* 104 (2021) 053006

More than two decades of NuWro

WroNG – Wrocław Neutrino Generator of events for single pion production

Jan T. Sobczyk^{a,*}, Jarosław A. Nowak^a, Krzysztof M. Graczyk^a

^aInstitute of Theoretical Physics, Wrocław University,
pl. M. Borna 9, 50-204 Wrocław, Poland

WroNG @ NuInt04



NuWro Workshop 2017

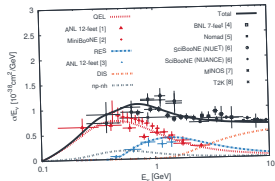
New models

User friendly

Wide applicability

Reliable results

Optimized code



- | | |
|--------------------------|--------------------------|
| [1] PRD 19 (1979) 2521 | [5] PLB 660 (2008) 19 |
| [2] PRD 81 (2010) 092005 | [6] PRD 83 (2011) 012005 |
| [3] PRD 16 (1977) 3103 | [7] PRD 81 (2011) 072002 |
| [4] PRD 25 (1982) 617 | [8] PRD 87 (2013) 092003 |

Kajetan Niewczas

NuWro @ NuInt17

June 28th, 2017

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NuWro @ NuInt17

Fine-tuning final state interactions model in NuWro Monte Carlo event generator

Hemant Prasad , Jan T. Sobczyk , Rwik Dharmapal Banerjee ,

J. Luis Bonilla , Krzysztof M. Graczyk , and Beata E. Kowal 
Institute of Theoretical Physics, University of Wrocław, Plac Maxa Borna 9, 50-204, Wrocław, Poland

Artur M. Ankowski 

M. Smoluchowski Institute of Physics, Faculty of Physics,
Astronomy, and Applied Computer Science, Jagiellonian University,
Ul. Profesora Stanisława Łojasiewicza 11, 30-348 Kraków, Poland
(Dated: December 30, 2025)

Latest NuWro improvements (2025)